Operating System Overview

Chapter 2

Layers of Computer System

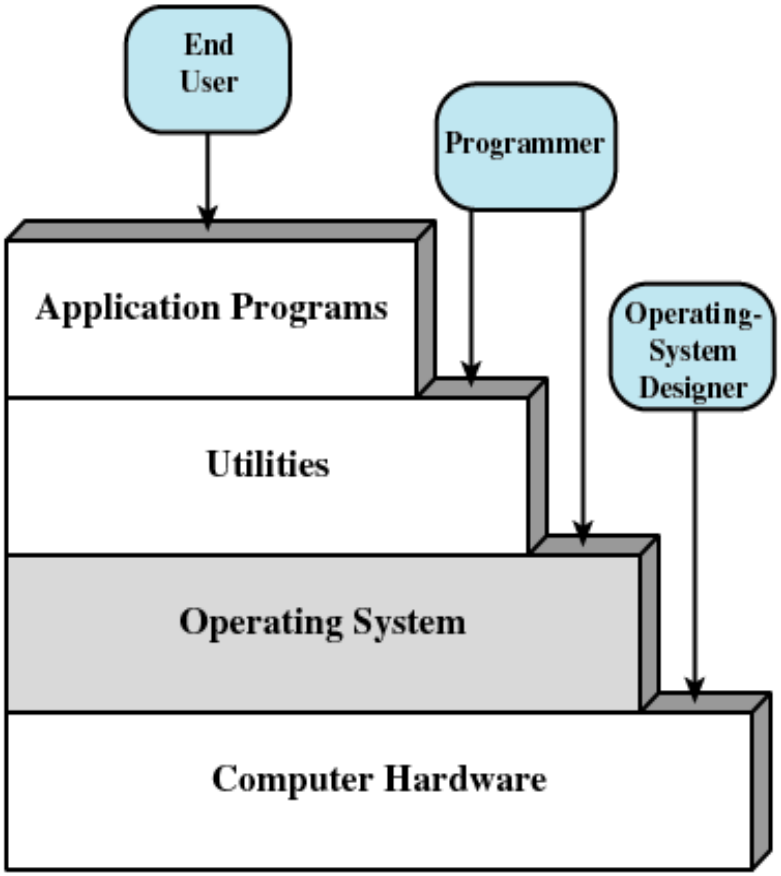


Figure 2.1 Layers and Views of a Computer System

Operating System Architecture

- A program that controls the execution of application programs
- An interface between applications and hardware

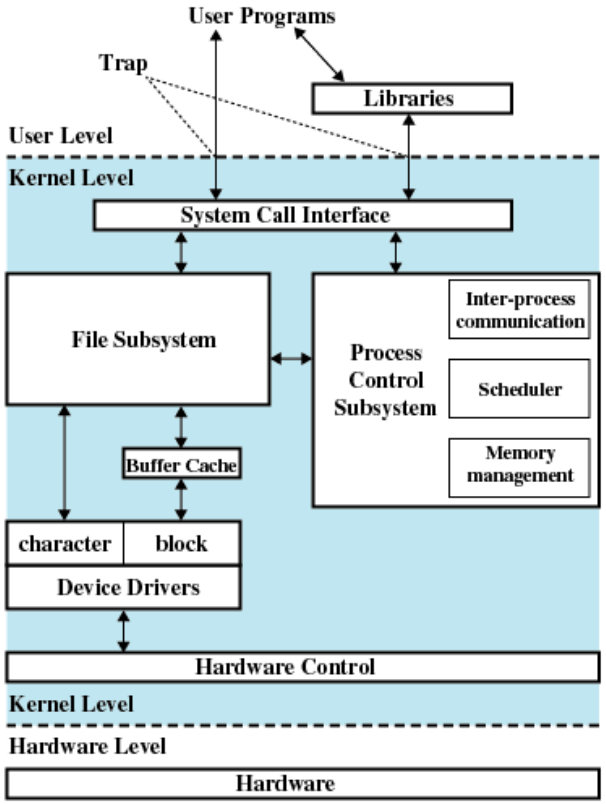


Figure 2.15 Traditional UNIX Kernel [BACH86]

Operating System Objectives

- Convenience
  - Makes the computer more convenient to use
- Efficiency
  - Allows computer system resources to be used in an efficient manner
- Ability to evolve
  - Permit effective development, testing, and introduction of new system functions without interfering with service

Services Provided by the OS

- Program development
  - Editors and debuggers
- Program execution
- Access to I/O devices
- Controlled access to files

Services Provided by the OS (cont.)

- Error detection and response
  - Internal and external hardware errors
    - Memory error
    - Device failure
  - Software errors
    - Arithmetic overflow
    - Access forbidden memory locations

Services Provided by the OS (cont.)

- Accounting
  - Collect usage statistics
  - Monitor performance
  - Used to anticipate future enhancements
  - Used for billing purposes

Kernel (核心/内核)

- Portion of operating system that is in main memory
- Contains most frequently used functions
- Also called the nucleus (核心)

Evolution of an OS

- Hardware upgrades plus new types of hardware
- New services
- Fixes

Evolution of OS (cont.)

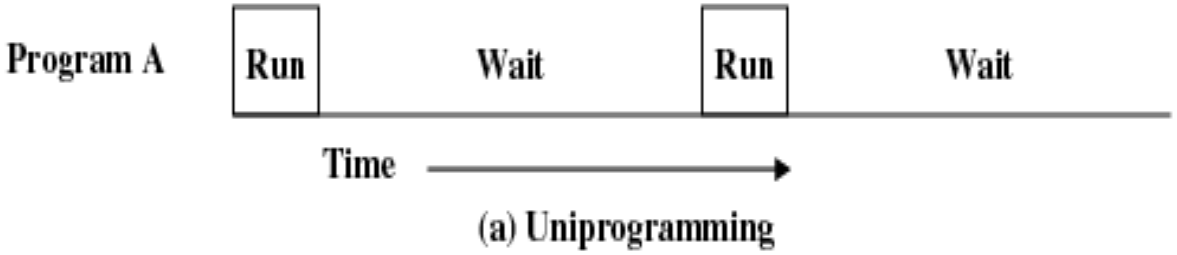
- Serial Processing
  - No operating system
  - Machines run from a console with display lights, toggle switches, input device, and printer
  - Schedule time
  - Setup included loading the compiler, source program, saving compiled program, and loading and linking

Evolution of OS (cont.)

- Simple Batch Systems
  - Monitors
    - Software that controls the sequence of events
    - Batch jobs together
    - Program branches back to monitor when finished
  - Job Control Language (JCL)
    - Special type of programming language
    - Provides instruction to the monitor

Evolution of OS (cont.)

- Uniprogramming (单任务)
- Processor must wait for I/O instruction to complete before preceding



(a) Uniprogramming

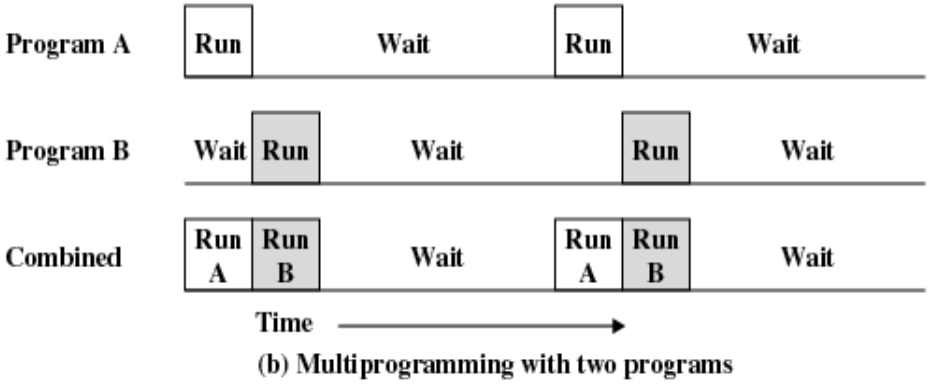
Read one record from file	15 μs
Execute 100 instructions	1 μs
Write one record to file	15 μs
TOTAL	31 μs

Percent CPU Utilization $= \frac{1}{31} = 0.032 = 3.2\%$

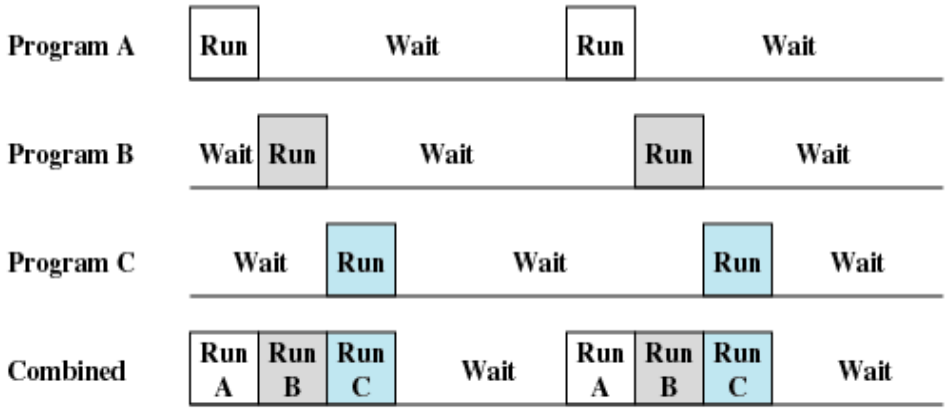
Figure 2.4 System Utilization Example

Evolution of OS (cont.)

- Multiprogramming (多道/多任务)
- When one job needs to wait for I/O, the processor can switch to the other job



(b) Multiprogramming with two programs



(c) Multiprogramming with three programs

Evolution of OS (cont.)

- Multiprogramming (多道/多任务)
  - Processor has more than one program to execute
  - The sequence the programs are executed depend on their relative priority and whether they are waiting for I/O
  - After an interrupt handler completes, control(控制) may not return to the program that was executing at the time of the interrupt
- Using multiprogramming(多道) to handle multiple interactive jobs, processor's time is shared among multiple users
- Multiple users simultaneously access the system through terminals

Evolution of OS (cont.)

- Time Sharing
- First time-sharing system developed at MIT

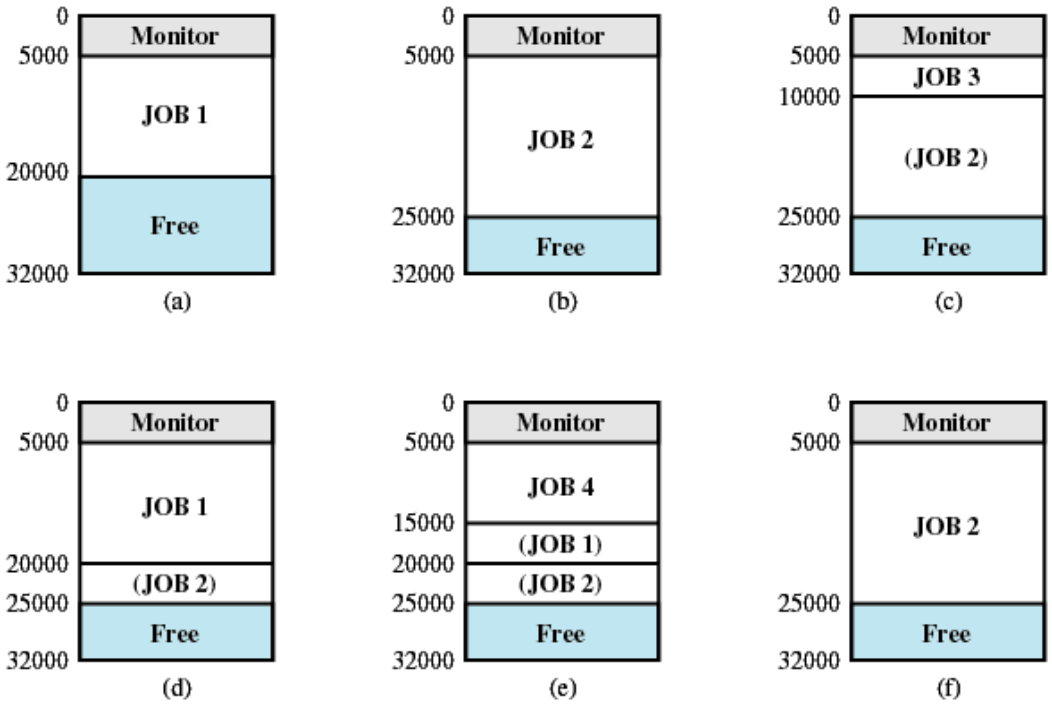


Figure 2.7 CTSS Operation

Features of Modern Operating Systems

- Microkernel architecture
  - Assigns only a few essential functions to the kernel
    - Address spaces
    - Interprocess communication (IPC)
    - Basic scheduling

Features of Modern Operating Systems

- Multithreading(多线程)
  - Process is divided into threads that can run concurrently
    - Thread
      - Dispatchable unit of work
      - executes sequentially and is interruptable
    - Process is a collection of one or more threads
- Symmetric multiprocessing (SMP)
  - There are multiple processors
  - These processors share same main memory and I/O facilities
  - All processors can perform the same functions

Features of Modern Operating Systems

- Multiprogramming(多道) and multiprocessing (多处理)

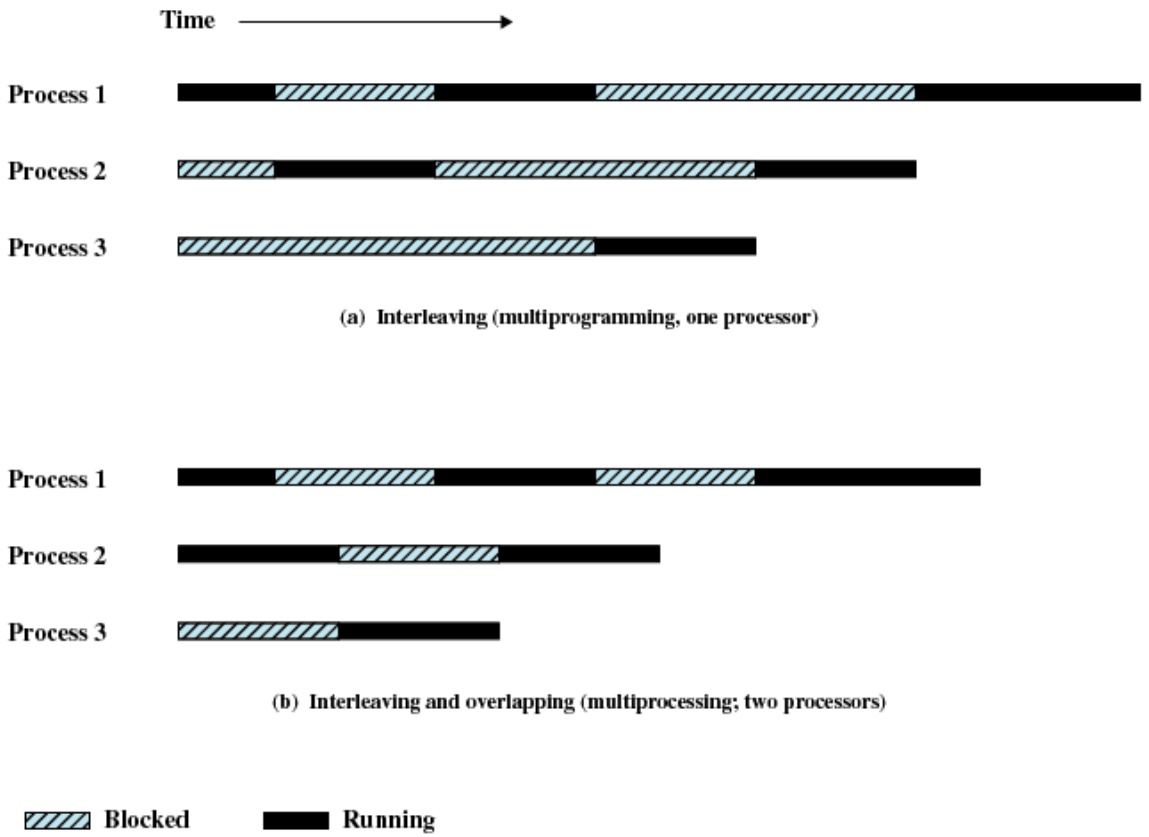


Figure 2.12 Multiprogramming and Multiprocessing

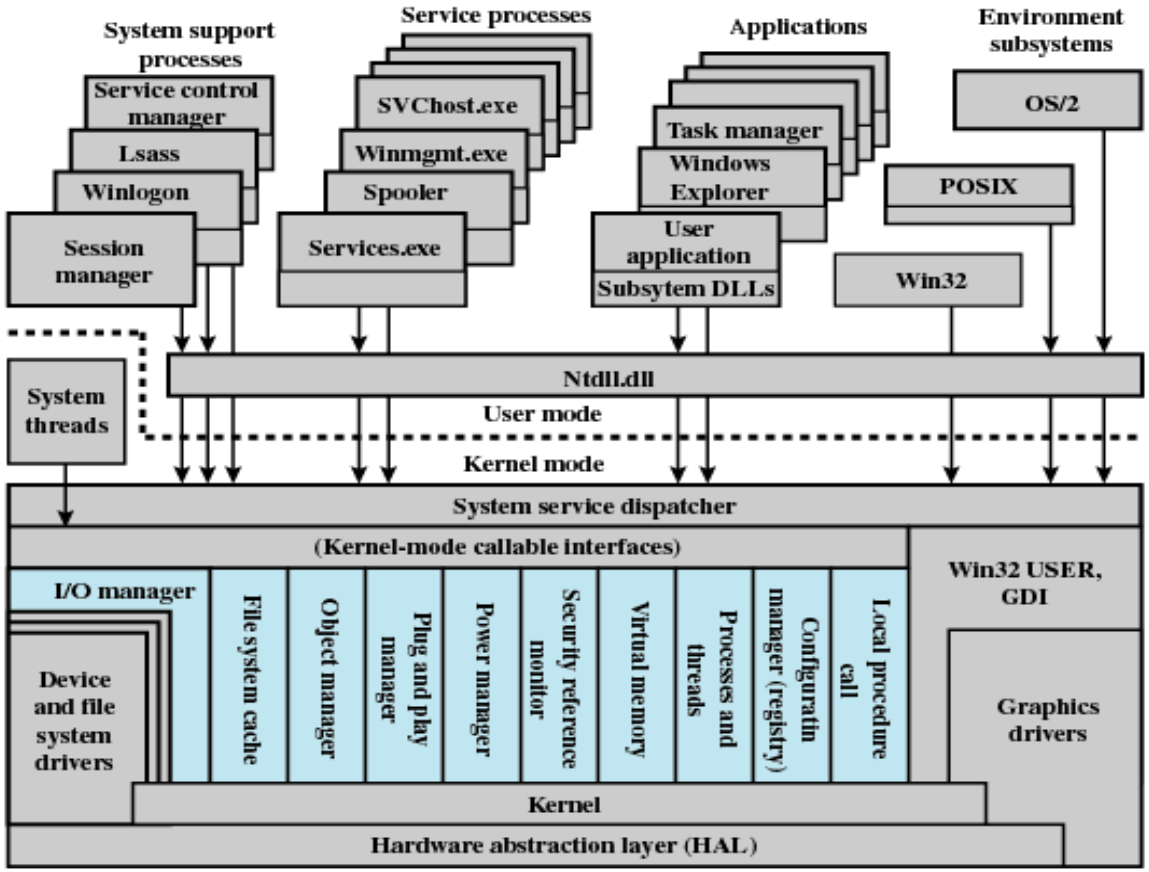
Features of Modern Operating Systems

- Distributed operating systems
  - Provides the illusion of a single main memory space and single secondary memory space
- Object-oriented design
  - Used for adding modular extensions to a small kernel
  - Enables programmers to customize an operating system without disrupting system integrity

Windows Architecture

- Modular structure for flexibility
- Executes on a variety of hardware platforms
- Supports application written for other operating system

Windows 2000 Architecture



Lsass = local security authentication server
POSIX = portable operating system interface
GDI = graphics device interface
DLL = dynamic link libraries

Colored area indicates Executive

Figure 2.13 Windows 2000 Architecture [SOLO00]

Traditional UNIX

- Hardware is surrounded by the operating system software
- Operating system is called the system kernel
- Comes with a number of user services and interfaces
  - Shell
  - Components of the C compiler

Traditional UNIX

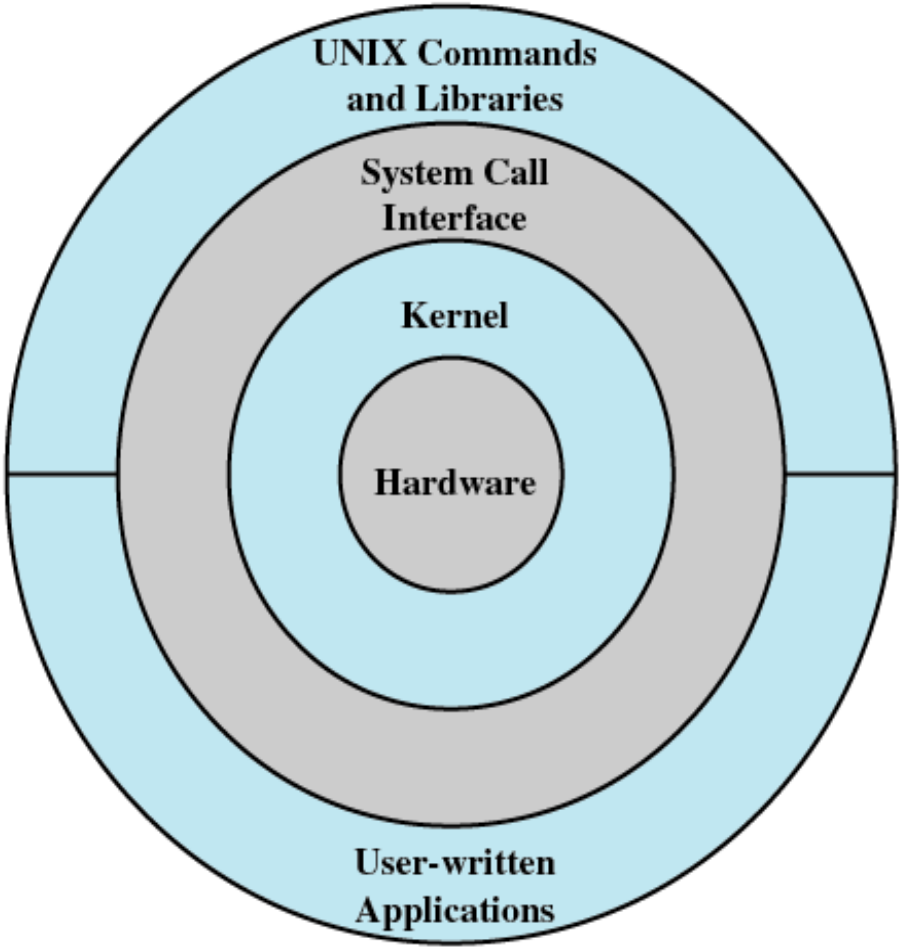


Figure 2.14 General UNIX Architecture

Traditional UNIX Kernel

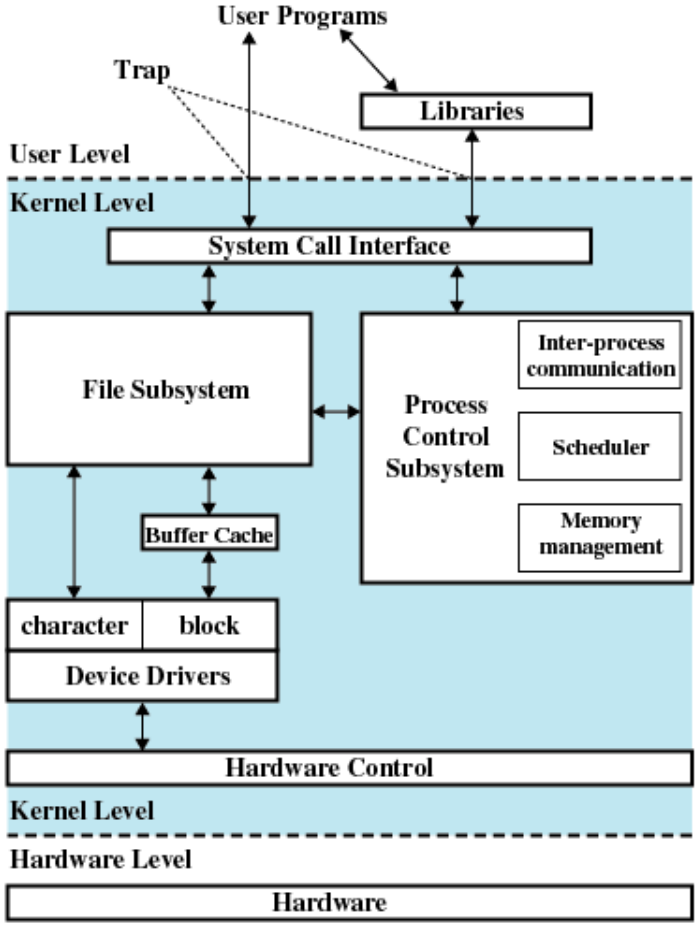


Figure 2.15 Traditional UNIX Kernel [BACH86]

Modern UNIX Kernel

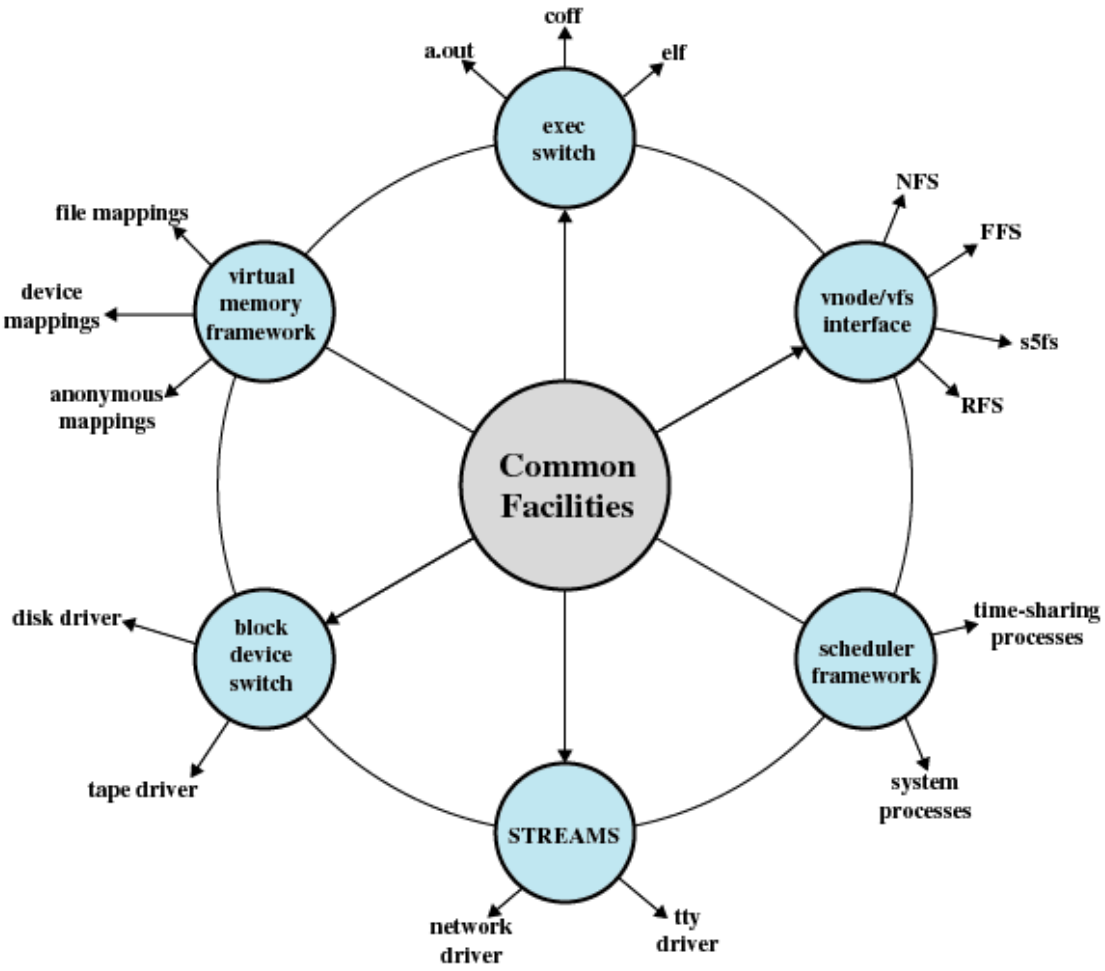


Figure 2.16 Modern UNIX Kernel [VAHA96]

Modern UNIX Systems

- System V Release 4 (SVR4)
- Solaris 9
- 4.4BSD
- Linux